Mixed Loss Graph Attention Network for Few-Shot SAR Target Classification

Minjia Yang, Xueru Bai, *Member, IEEE*, Li Wang, and Feng Zhou, *Member, IEEE*

***Abstract*— Restricted by the observation condition, synthetic aperture radar (SAR) automatic target classification based on deep learning usually suffers from insufficient training samples. To tackle this problem, a novel few-shot learning (FSL) framework for SAR target classification, i.e., the mixed loss graph attention network (MGA-Net), is proposed. The classification procedure of the MGA-Net consists of three main stages. In the first stage, the task set is expanded by the data augmentation module to increase diversity. In the second stage, the embedding network is designed to map samples to the embedding space with strong intra-class similarity and inter-class divergence. In the third stage, the multilayer graph attention network (GAT) is constructed and updated according to a novel mixed loss to obtain the classification result. In particular, the data augmentation module alleviates the desire of training samples under large model capacity and enhances the robustness to noise and viewing angle variation; the multilayer GAT accurately captures relations between samples by the attention mechanism; and the mixed loss increases the inter-class separability and accelerates convergence. Experimental results under various few-shot observation settings of the MSTAR and the OpenSARShip benchmark datasets demonstrate that the MGA-Net obtains higher accuracy than typical FSL methods and exhibits robustness to large depression angle variation.**

***Index Terms*— Automatic target classification, deep learning, few-shot learning (FSL), graph attention network (GAT), synthetic aperture radar (SAR).**

I. Introduction

THANKS to its all-weather, day-and-night, long-distance, and high-resolution capabilities, synthetic aperture radar (SAR) has played a significant role in geological observation, military reconnaissance, and battlefield perception [1]. Compared with the optical image, the gray-scale, edge-blurred, single-polarized SAR image is anisotropic and prone to be interfered by strong background clutter, challenging the available automatic target classification techniques [2]–[34].

Traditional classification methods [2]–[9] depend heavily on hand-designed features, which require a large amount of expertise. In addition, they usually have heavy computational burden and poor generalization performance to new categories. With the rapid development of machine learning techniques, the convolutional neural network (CNN) has become an extremely popular and indispensable architecture in SAR target classification and outperforms traditional approaches by a large margin. As a salient property, the data-driven mechanism avoids the tedious and complexity of feature extractor design. In addition, the local connections, shared weights, pooling, and the use of many layers [35] facilitate deeper networks and hierarchical feature representations, making major advances in solving problems of SAR target feature extraction and classification [10]–[34].

In contrast, deep CNN usually has huge amounts of parameters (possibly millions) and requires abundant, high-quality training samples to alleviate overfitting. In practice, however, SAR data collection and sample-labeling can be quite expensive and painstaking, therefore, abundant samples are unavailable, and some target categories may have only a few (several or dozens) labeled samples. In this scenario, the performance of CNN degrades significantly and the problem of few-shot SAR target classification arises.

Recent studies on few-shot learning (FSL) [36] may shed light on the hard yet significant problem. Mimicking the human ability to acquire knowledge from few examples through generalization, FSL first trains the model with sufficient samples of common categories, then extends the learned classification capability to new categories with limited labeled samples, and finally classifies the unlabeled samples by the supervision information of labeled ones. In view of this, the task-driven FSL generates the training tasks $\mathcal{T}_{\text{train}}$ and the tests tasks $\mathcal{T}_{\text{test}}$ by respective and repeated sampling from the training set (which includes common categories) and the test set (which includes new categories) and simulates the real few-shot classification tasks in $\mathcal{T}_{\text{test}}$ by training on $\mathcal{T}_{\text{train}}$ to learn the classification capability. Specifically, in each task $\mathcal{T}$, the labeled samples constitute the support set $\mathcal{S}$ and the unlabeled ones constitute the query set $\mathcal{Q}$. If there are C unique target types in $\mathcal{S}$ and each one contains K samples, then the task is dubbed the C-way K-shot problem, as illustrated in Fig. 1.

A. Prior Work

The success of feature learning is dependent on efficient and robust network structure. Characterized by the superior feature extraction capability on Euclidean data (e.g., the SAR image), CNN has become a mainstream feature extractor in SAR automatic target classification. Recently, its variants have

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Fig. 1. Training and test tasks of the 3-way 1-shot setup, where each row represents a task, and pictures share the same color border in each task belonging to the same category.

achieved improved classification performance by: deleting the fully connected layer [10], constructing the multichannel structure [11], adopting batch normalization [12], adding the information recorder [13], constructing the cascade network [14], and introducing the attention mechanism [15]. Meanwhile, multifeature fusion networks are designed to exploit the geometric, spatial, temporal, and other significant features. For instance, the spatial-temporal features of the sequence SAR images are captured by the bidirectional convolutional recurrent network (BCRN) [16] and the spatial-temporal ensemble convolutional network (STEC-Net) [17], while multimode fusion features are obtained by CNN and sparse coding [18]. In addition, the autoencoder (AE) compresses information of the SAR image by the encode–decode structure and treats codes in the middle layer as features [19]–[21]. However, due to the incompetence in geometric structure extraction caused by the simplicity of network structure and deformation of the input data, these AE methods have limited feature representation capability.

Traditionally, deep CNNs usually require a large amount of training data to avoid overfitting. Restricted by the observation condition as well as the expensive and time-consuming manual annotation, however, the training data are insufficient for SAR automatic target classification. To tackle this issue, the existing methods mainly: 1) augment the training set by image cropping, shifting, rotating, and distorting [22]–[24]; 2) train the network by semisupervised learning [25]–[28] or transfer learning [29]–[32]; and 3) modify the network structure by reasonably reducing the number of parameters [10] or design-specific networks based on the domain knowledge of SAR image [33]–[34]. Specifically, semisupervised learning combines labeled and unlabeled samples for network training, while transfer learning utilizes sufficient samples in the source domain to pretrain the model and then applies the small amount of samples in the target domain for fine-tuning. To obtain robust classification performance, the above techniques still require at least dozens or hundreds of training samples for each target category.

According to the theory of FSL, embedding learning [37]–[42] maps the samples of the original space to a lower dimensional embedding space by the embedding network, such that similar samples are close together, while dissimilar samples can be more easily differentiated. Then, the similarity between samples in the embedding space is measured by metrics such as the cosine similarity [38], the Euclidean distance [39], the Gaussian similarity [40], and the learned distance [41]–[42] to make a classification decision. For classical embedding learning methods, the prototypical network (ProtoNet) [39] synthesizes information of similar samples by calculating the prototype center of each class in the embedding space and realizes few-shot classification by only comparing the Euclidean distance between the unlabeled samples and prototype centers. In contrast, the relation network (RelationNet) [41] and the attention metric network (AMN) [42] construct learnable inference networks. For SAR target classification, however, features of the same category may change greatly due to strong background clutter and variation of the viewing angle. In this case, it is difficult to generalize the clustering characteristics of the training set to the test set, and the distribution of embedding vectors within the same category exhibits various manifold structures in the embedding space. Because the ProtoNet, the RelationNet, and AMN depend heavily on the similarity between unlabeled and labeled samples while ignoring the global manifold in the embedding space, their performance may degrade heavily with inaccurate description of the sample relations.

Featured by the strong capability in non-Euclidean data modeling and the flexibly of information propagation, the graph neural network (GNN) [43] can effectively represent the complex relations among samples and describe the global manifold structure of their distribution; thus, it avoids the irrationality of directly utilizing the distance measurement. Specifically, for a fully connected graph composed of nodes and edges, the node positions are presented by node features, and the dependency between nodes is obtained by aggregating and updating the node feature multiple times. In addition, GNN is learnable, i.e., it adjusts the metric strategy in embedding learning according to the input data flexibly, and improves the performance of FSL [44]–[46]. Still, it will encounter the following issues.

1) Typically, GNN constructs all the samples into a fully connected graph, where each node acquires information from all other nodes during node feature aggregation. Such setting hinders effective information enhancement and makes the nodes vulnerable to the information flow of unrelated ones. Accordingly, the differentiation of node features is reduced in deep layers, i.e., the GNN tends to be over-smooth [47], after multiple iterations.
2) When adopted for FSL, GNN is prone to overfitting and slow convergence in the training stage, owning to the large number of network parameters and limited SAR image samples of common target categories in the training set.

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Widely applied in sequence learning tasks, the attention mechanism allows for focusing on the most relevant parts of the input to make decisions. By introducing the attention mechanism to GNN, the graph attention network (GAT) [48] constructs the attention weight between nodes to enhance the effective information during node feature aggregation and improves the dissimilarity of node features among different target categories. For few-shot SAR automatic target classification, however, GAT still suffers from large number of model parameters, limited number of training samples, and slow convergence.

B. Contribution

To tackle the issues mentioned above, this article proposes a novel framework for few-shot SAR target classification, and the contributions can be summarized as follows.

1) The Mixed loss Graph Attention Network (MGA-Net) is proposed for few-shot SAR target classification, which consists of the data augmentation module, the embedding network, and GAT. Compared with classical FSL methods, the MGA-Net achieves higher average classification accuracy and exhibits more robustness in various experiments on the MSTAR and the OpenSARShip benchmark datasets.
2) To tackle the issues of large depression angle sensitivity and feature variation, the inference network based on multilayer GAT is designed by constructing a graph model of samples in the embedding space. Featured by the strong flexibility and non-Euclidean structure, it adapts to large feature changes within the same category and avoids the decrease of separability among categories. In addition, it enhances the effective information during node feature aggregation and alleviates over-smooth in deep layers of GNN.
3) To tackle the issues of blurred edges and strong background clutter in SAR images, the data augmentation module is designed, which includes the power transformation, noise adding, flipping, and rotating operations. By simulating the image variations in practice, it improves the model robustness and prevents overfitting during training.
4) A novel mixed loss, i.e., the weighted sum of the embedding loss and classification loss, is designed, which accelerates convergence, improves the intra-class similarity and inter-class dissimilarity of samples in the embedding space, and provides solid support for the classification based on GAT.

II. Network Structure

This section introduces the structure of MGA-Net, which consists of three modules, namely the data augmentation module, the embedding network, and GAT. First, the data augmentation module implements affine transformations on the input SAR images for sample expansion. Then, the embedding network maps the samples into the embedding space and constructs the initial node features. Finally, the GAT module calculates the multihead attention weights, updates the edge and node features in each layer, and outputs the classification results of unlabeled nodes. The framework of MGA-Net is shown in Fig. 2, and detailed structures will be introduced in Section II-A–II-D.

A. Data Augmentation Module

In FSL, a large scale $\mathcal{T}_{\text{train}}$ is desired for effective network training. In addition, blurred edge, background clutter, and the variation of target orientation usually exist in SAR images. To increase the diversity of the support set in the training stage and gain robustness to noise and pose variation, the data augmentation module expands the training set via image intensity scaling (by power transform), distorting (by adding noise, flipping left to right, and turning upside down), and rotating (by 90°, 180°, and 270°), as shown in Fig. 3. Specifically, the power transform is expressed as $(I_{\text{in}}/255)^{\gamma} \times 255$ with $\gamma \in [0.7, 1.3]$. For $\gamma < 1$, small pixel values will be enhanced, while large pixel values will be suppressed and vice versa for $\gamma > 1$. By this means, the network acquires geometry and structural features with strong and weak scattering intensities from the original image I_{in}. The added noise obeys the uniform distribution. In addition, the flipping, turning, and rotating operations are carried out randomly with equal probabilities, which increase the sample diversity while preventing the loss of original features.

B. Embedding Network

The embedding network maps the augmented SAR images to low-dimensional embedding vectors in the embedding space with strong intra-class similarity and inter-class dissimilarity. For the ith sample x_i, the embedding vector $\mathbf{z}_i$ satisfies

$$\mathbf{z}_i = f_\theta(x_i) \tag{1}$$

where $f_\theta : \mathbb{R}^{H\times W} \rightarrow \mathbb{R}^P$ is the embedding function, θ is learnable parameter set, H and W are the height and width of the input image respectively, and P is the dimension of $\mathbf{z}_i$.

Although a variety of embedding functions are adopted in literature, a CNN without fully connected layers is utilized to reduce the model parameters. As shown in Fig. 4, the first five convolutional blocks (ConvBlock) include convolutional layers and max-pooling layers, while the last convolutional block (ConvLast) only includes a convolutional layer to compress the feature maps to embedding vectors. By "Conv64 @ K3_S1_P1/BN/Mish," we mean that the convolutional layer has 3 3 $\times$ 3 kernels with step size 1 and padding size 1, followed by batch normalization (BN). In addition, we adopt the Mish activation function [49], which obtains better performance than ReLU/leaky ReLU [50] and Swish [51]. Besides, "MaxPooling@K2_S2" indicates max-pooling with kernel size 2×2 and step size 2.

To construct the initial node features $\mathbf{x}_i^{(1)} \in \mathbb{R}^{d_1}$ of the ith node in GAT, we concatenate the embedding vector of the input image and its label, i.e.

$$\mathbf{x}_i^{(1)} = \left[f_\theta(x_i)^{\mathrm{T}}, h(y_i)^{\mathrm{T}}\right]^{\mathrm{T}} \tag{2}$$

where the superscript of the initial node features $\mathbf{x}_i^{(1)}$ indicates the first layer of the GAT; the superscript of $\mathbb{R}^{d_1}$ represents the

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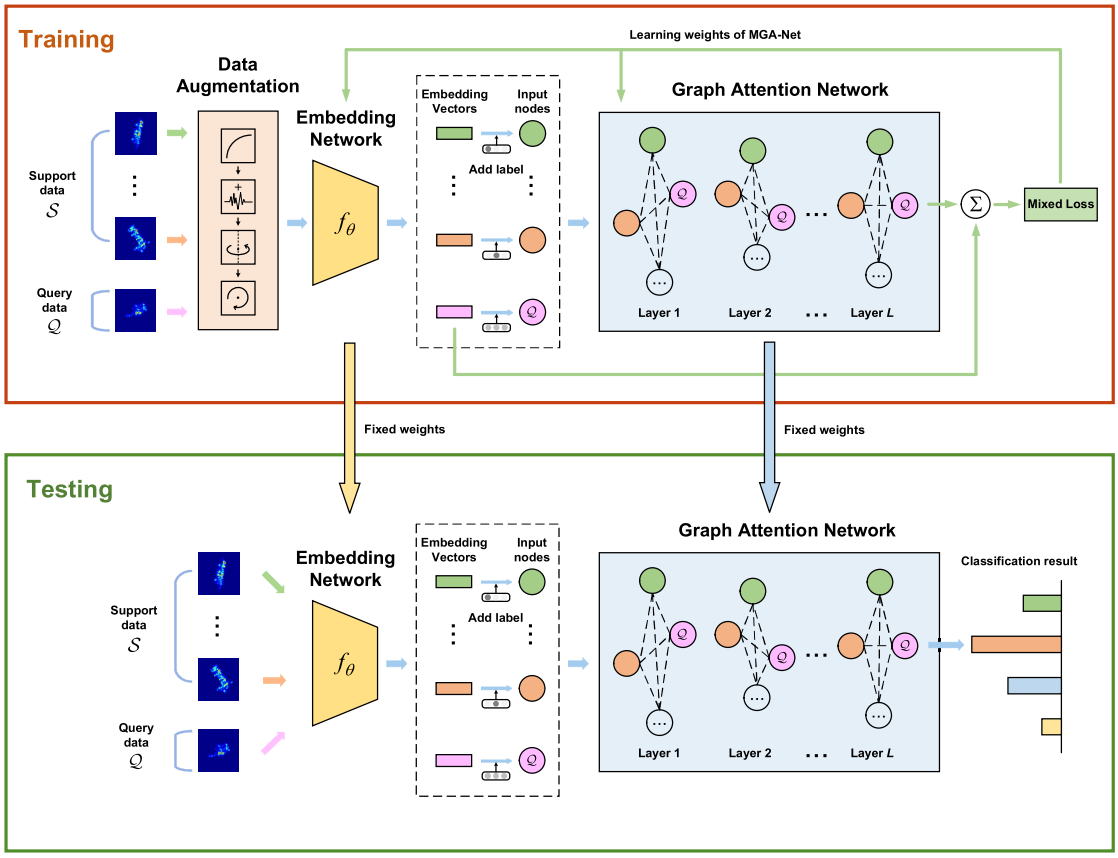


Fig. 2. Framework of the MGA-Net.

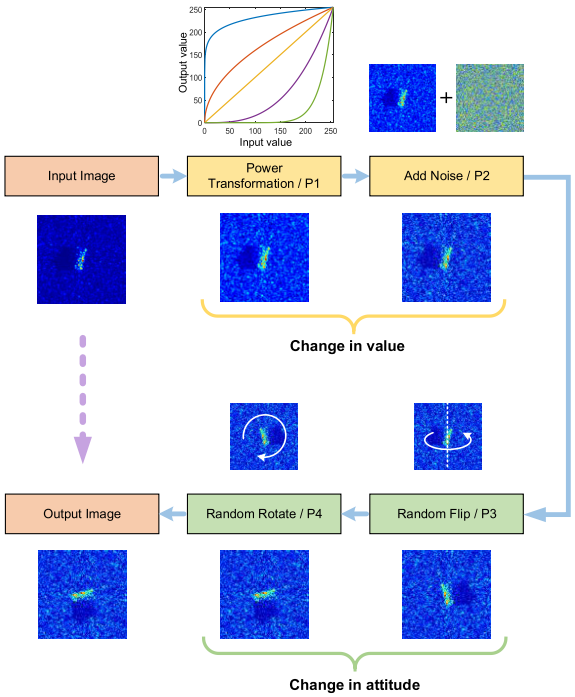


Fig. 3. Illustration of the data augmentation module.

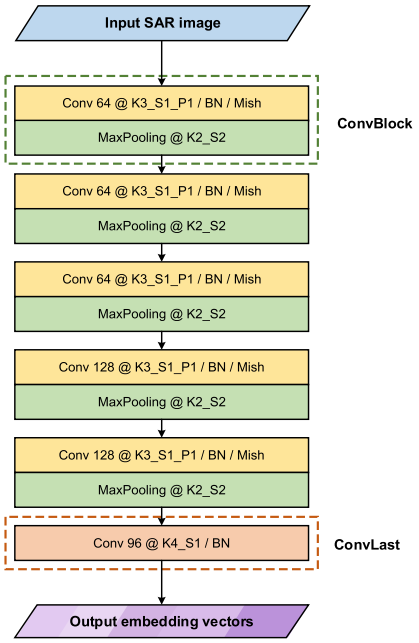


Fig. 4. Detailed structure of the embedding network.

dimension of $\mathbf{x}_i^{(1)}$; y_i is the label of the ith sample; $h(y_i) \in \mathbb{N}^C$ is the one-hot coding of y_i; C is the number of target categories in the support set; and $(\cdot)^{\mathrm{T}}$ is vector transpose. For the query sample to be classified, we set $h(y) = C^{-1}\mathbf{1}_C$, where $\mathbf{1}_C$ is a C-dimension vector with all ones.

C. Graph Attention Network (GAT)

In order to effectively capture the global manifold structure of sample distribution in the embedding space, we design

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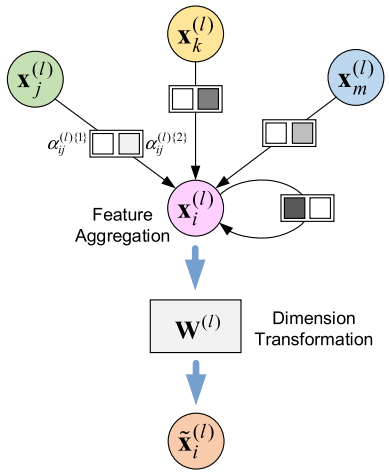


Fig. 5. Illustration of node feature update with multihead attention $H = 2$.

the learnable multilayer multihead GAT, which serves as the metric module in embedding learning and outputs the classification results. Specifically, GAT takes the initial node features $\mathbf{x}_i^{(1)}$ as input, updates node features by the attention mechanism, and outputs labels of the query nodes.

For task $\mathcal{T}$, the graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}; \mathcal{T})$ is constructed by nodes $\mathcal{V} = \{V_i\}_{i=1}^{M}$ and edges $\mathcal{E} = \{E_{ij}\}_{i,j=1}^{M}$, $(i \neq j)$, where V_i is the ith node, E_{ij} is the edge from node V_i to node V_j, and M is the number of samples in $\mathcal{T}$. In MGA-Net, we adopt the fully connected graph, i.e., connection exists between arbitrary two nodes for $i \neq j$. The node feature of V_i and edge feature of E_{ij} in the lth ($l \in [1, L]$) layer are denoted by $\mathbf{x}_i^{(l)} \in \mathbb{R}^{d_l}$ and $e_{ij}^{(l)} \in \mathbb{R}$, respectively.

Each layer of GAT includes three parts: edge feature update, attention weight calculation, and node feature update. The edge feature $e_{ij}^{(l)}$ in the lth layer is given by a learnable function $a_\theta^{(l)}$, i.e.

$$e_{ij}^{(l)} = a_\theta^{(l)}\left(\mathbf{x}_i^{(l)}, \mathbf{x}_j^{(l)}\right) \tag{3}$$

where $a_\theta^{(l)}$ is a multilayer perceptron whose input is the absolute difference between the individual features of V_i and V_j, i.e.

$$a_\theta^{(l)}\left(\mathbf{x}_i^{(l)}, \mathbf{x}_j^{(l)}\right) = \mathrm{MLP}_\theta^{(l)}\left(\left|\mathbf{x}_i^{(l)} - \mathbf{x}_j^{(l)}\right|\right). \tag{4}$$

Accordingly, $a_\theta^{(l)}(\mathbf{x}_i^{(l)}, \mathbf{x}_j^{(l)}) = a_\theta^{(l)}(\mathbf{x}_j^{(l)}, \mathbf{x}_i^{(l)})$ and the feature symmetry property for E_{ij} and E_{ji} is fulfilled.

In order to enhance the effective information during node feature aggregation, GAT adopts the attention mechanism between nodes. Specifically, the proportion of information received by node V_i from V_j is determined by the attention weight, which is the output of a softmax function

$$a_{ij}^{(l)} = \mathbf{A}_{i,j}^{(l)} = \mathrm{softmax}_j\left(e_{ij}^{(l)}\right) = \frac{\exp\left(e_{ij}^{(l)}\right)}{\sum_{n \in \mathcal{N}_i} \exp\left(e_{in}^{(l)}\right)} \tag{5}$$

where $\mathbf{A}^{(l)} \in \mathbb{R}^{M \times M}$ is the attention matrix in the lth layer, and $\mathcal{N}_i$ is the neighbor of V_i, i.e., $\{V_j\}_{j \neq i}$. Since there is no connection between nodes for $i = j$, $\mathbf{A}_{i,j}^{(l)}(i = j) = 0$ holds.

To allow the model to jointly attend to information from different representation subspaces at different positions, multihead (H-head) attention is introduced through $\mathcal{A}^{(l)} = \{\mathbf{A}^{(l)\{h\}}\}_{h=1,\ldots,H}$. In addition, we set $\mathbf{A}^{(l)\{1\}} = \mathbf{I}$, where $\mathbf{I}$ represents the identity matrix. Therefore, the self-attention weight of each node equals to 1 and the mutual-attention weight of each node equals to 0. By this means, we prevent the loss of original features during node feature update.

Then, the node features are updated by

$$\tilde{\mathbf{x}}_i^{(l)} = \left[\Big\Vert_{h=1}^{H}\left(\sum_{j=1}^{M} \mathbf{A}_{i,j}^{(l)\{h\}}\, \mathbf{x}_j^{(l)}\right)\right] \mathbf{W}_\theta^{(l)} \tag{6}$$

where $\Vert$ denotes vector concatenation, which integrates multiple features of multihead attention, $\sum_{j=1}^{M} \mathbf{A}_{i,j}^{(l)\{h\}}\, \mathbf{x}_j^{(l)}$ denotes node feature aggregation, i.e., the information propagation between nodes weighted by the hth attention matrix, and $\mathbf{W}_\theta^{(l)}$ is a learnable parameter matrix designed for dimension transformation. For $h = 1$, we obtain the original node feature of V_i; for $h \neq 1$, we obtain the weighted sum of features of V_i's neighbor nodes. The schematic is shown in Fig. 5.

To further mitigate the over-smooth phenomenon in deep layers of GAT, we concatenate $\tilde{\mathbf{x}}_i^{(l)}$ and the original node feature as $\mathbf{x}_i^{(l+1)}$, and feed it into the next layer, i.e.

$$\mathbf{x}_i^{(l+1)} = \tilde{\mathbf{x}}_i^{(l)} \Vert \mathbf{x}_i^{(l)}. \tag{7}$$

For the Lth layer, we obtain the C-dimension feature vector $\tilde{\mathbf{x}}_i^{(L)}$ of the query sample, where each entry corresponds to the probability that the sample belongs to. Considering the relatively small training dataset (usually hundreds of samples) in automatic SAR target classification, we set $L = 3$ and $H = 2$.

D. Construction of the Mixed Loss

Due to the large amount of model parameters in GAT, problems such as slow convergence and long training time arise and degrade the classification performance. In view of this, we introduce the embedding loss, which acts solely on the embedding network and directly guides the learning of the embedding space. Meanwhile, the mixed loss, which consists of the classification loss and the embedding loss, is designed for the training of MGA-Net. By this means, we reduce the complexity of optimal model searching and speed up convergence.

In order to achieve strong intra-class similarity and inter-class divergence in the embedding space, we construct the embedding loss from the average embedding vector of the support set. The cluster center of the nth category is

$$\mathbf{c}_n = \frac{1}{K} \sum_{\mathbf{z}_i \in \mathcal{S}^{(n)}} \mathbf{z}_i \tag{8}$$

where $\mathcal{S}^{(n)}$ is the support set of the nth category, $\mathrm{card}\{\mathcal{S}^{(n)}\} = K$ (where $\mathrm{card}\{A\}$ denotes the cardinal number, i.e., the number of elements, in set A), and $\mathbf{z}_i$ is the embedding vector of the ith support sample. Then, the probability that the

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embedding vector of the query sample $\mathbf{z}_i$ belongs to label $\hat{y}_i$, i.e., $p(\hat{y}_i = n|\mathbf{z}_i)$, is given by the softmax function

$$p(\hat{y}_i = n|\mathbf{z}_i) = \frac{\exp(-d(\mathbf{z}_i, \mathbf{c}_n))}{\sum_{j=1}^{C} \exp\left(-d\left(\mathbf{z}_i, \mathbf{c}_j\right)\right)} \tag{9}$$

where d is the metric function and satisfies $d(\mathbf{z}_i, \mathbf{c}_n) = ||\mathbf{z}_i - \mathbf{c}_n||_2^2$.

Finally, the embedding loss L_E is constructed by the negative logarithm probability

$$L_E = -\frac{1}{Q} \sum_{i=1}^{Q} \log\, p(\hat{y}_i = y_i|\mathbf{z}_i) \tag{10}$$

where Q is the number of query samples.

The classification loss compares the difference between predicted labels and true ones. For the ith query sample, the last layer of GAT generates the node feature $\tilde{\mathbf{x}}_i^{(L)} \in \mathbb{R}^C$ and utilizes the softmax function to calculate the probability that the sample belongs to the jth class $\hat{\mathbf{y}}_{i,j} \in \mathbb{R}^C$. Then, the cross-entropy loss L_C is constructed by

$$L_C = -\frac{1}{Q} \sum_{i=1}^{Q} \sum_{j=1}^{C} \mathbf{y}_{i,j} \log\left(\hat{\mathbf{y}}_{i,j}\right). \tag{11}$$

In MGA-Net, the learnable parameters exist in the embedding network f_θ, the multilayer perceptron MLP_θ in edge feature update, and the dimensional transformation matrix $\mathbf{W}_\theta$ in node feature update. Denoting the corresponding parameter sets by θ_f, θ_m, and θ_w, respectively, then, the entire parameter set for the whole model is $\Theta = \{\theta_f, \theta_m, \theta_w\}$. By definition, L_C acts on Θ, while L_E only acts on θ_f.

Finally, the mixed loss L is the weighted sum of the embedding loss and the classification loss

$$L(\Theta) = \lambda L_C(\Theta) + (1 - \lambda) L_E\left(\theta_f\right) \tag{12}$$

where λ $(0 \le \lambda \le 1)$ is the weight.

In the training stage, back-propagation is applied and Θ is updated along the negative gradient direction of $L(\Theta)$, i.e.

$$\Theta_{\text{new}} = \Theta_{\text{old}} - \eta \nabla L(\Theta_{\text{old}}) \tag{13}$$

where Θ_{old} is the parameter set before update, Θ_{new} is the parameter set after update, η is the learning rate, and $\nabla L(\Theta_{\text{old}})$ is the partial derivative of the loss with respect to Θ_{old}, i.e., $\nabla L(\Theta_{\text{old}}) = \partial L / \partial\, \Theta_{\text{old}}$.

III. Experiment

A. Description of the MSTAR Dataset

In this section, we adopt the MSTAR benchmark dataset [52] provided by the Defense Advanced Research Projects Agency (DARPA)/Air Force Research Laboratory (AFRL) to design few-shot classification experiments for network validation and performance analysis.

The dataset was collected by X-band HH-polarization spotlight SAR with resolution of 0.3 m × 0.3 m. The image size is 128 × 128; the depression angles are 15°, 17°, and 30°; and the azimuth angle changes continuously from 0° to 360° with an interval of about 5°. The dataset includes ten categories of ground military vehicle targets, i.e., 2S1, BMP-2, BRDM-2, BTR-60, BTR-70, D-7, T-62, T-72, ZIL-131, and ZSU-234. The typical SAR images and the corresponding optical images are shown in Fig. 6.

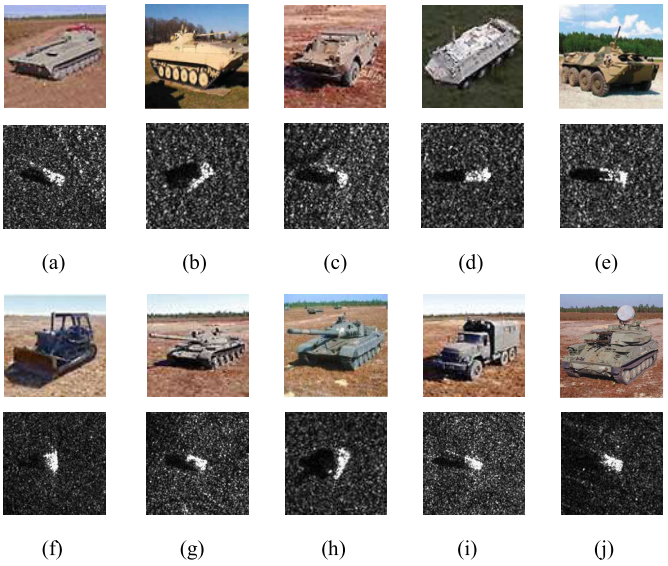


Fig. 6. Ten categories of targets in the MSTAR dataset. (Top) Optical images versus (Bottom) SAR images. (a) 2S1. (b) BMP-2. (c) BRDM-2. (d) BTR-60. (e) BTR-70. (f) D-7. (g) T-62. (h) T-72. (i) ZIL-131. (j) ZSU-234.

TABLE I
DIVISION OF THE MSTAR DATASET

	Target Class	Depression	Sample Number
Training set	BMP-2	17°	200
	BTR-70		
	T-72		
	BTR-60		
	D-7		
	T-62		
	ZIL-131		
Test set	2S1	15°	274
	BRDM-2	17°	298
	ZSU-234	30°	287

To construct the training task set $\mathcal{T}_{\text{train}}$ and the test task set $\mathcal{T}_{\text{test}}$, we divide the original dataset into disjoint training set $\mathcal{D}_{\text{train}}$ and test set $\mathcal{D}_{\text{test}}$ (as shown in Table I) based on the following reasons.

1) *Depression Angle:* In the MSTAR dataset, all categories contain samples with depression angle of 17°, while only three categories, i.e., 2S1, BRDM-2, and ZSU-234 contain samples with depression angles of 15°, 17°, and 30° simultaneously. In order to verify the model robustness to large depression angle variation, we set the depression angle of the seven categories in the training set to 17°; and set the depression angles of the three categories in the test set to 15°, 17°, and 30°, respectively.

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TABLE II
EXPERIMENT SETTINGS OF THE MSTAR DATASET

No.	Experiment type	Depression		
		Training tasks	Test tasks	
			Support set	*Query set*
1	15s/15q	17°	15°	15°
2	17s/15q		17°	15°
3	17s/17q		17°	17°
4	30s/30q		30°	30°
5	17s/30q		17°	30°

2) *Sample Number:* Because the number of samples differs among categories in the MSTAR dataset, we randomly select 200 samples for each category in the training set to guarantee the balance of samples and prevent overfitting. In order to comprehensively evaluate the model performance, we select all the samples of 2S1, BRDM-2, and ZSU-234 to form the test set.

Finally, as a matter of routine [39], we will carry out 3-way 1-shot and 3-way 5-shot experiments in the following discussions.

B. Training and Testing Process

Most existing FSL algorithms based on deep learning adopt the episode-based training method [37]–[39]. The main idea is to simulate the test stage during training. In view of this, the training and test tasks in $\mathcal{T}_{\text{train}}$ and $\mathcal{T}_{\text{test}}$ share the same C-way K-shot problem.

To construct $\mathcal{T}_{\text{train}}$, we first select C arbitrary categories from $\mathcal{D}_{\text{train}}$ and then pick K supervised samples randomly from each category to form the support set $\mathcal{S}$; then, we choose 1 sample randomly from the remaining samples to form the query set $\mathcal{Q}$. The above process is repeated T times, i.e., there are T tasks in $\mathcal{T}_{\text{train}}$.

To construct the test task set $\mathcal{T}_{\text{test}}$, we utilize five types of support-query combinations, where the depression angles may vary between the support set and the query set, as shown in Table II. First, we generate K support samples for each category by random sampling of the test set at the specified depression angle. Then, we generate the query set as follows.

Case 1: The depression angle of the query set differs from that of the support set, e.g., 17 s/15 q. Select the query samples of each category at the specified depression angle one by one. Then, form the test tasks by combining the support samples and each query sample, respectively. In this case, each query set has only one sample, and the number of test tasks equals to the total number of samples for the corresponding depression angle.

Case 2: The query set and the support set share the same depression angle, e.g., 15 s/15 q. At the specified depression angle, select the query samples from the remaining samples of each category one by one. Then, form the test tasks by combining the support samples and each query sample, respectively. Similarly, each query set has only one sample, but the number of test tasks is the total number of samples in the corresponding depression angle minus the total number of support samples.

TABLE III

Algorithm 1 Training process of the MGA-Net

Input: Training set $\mathcal{D}_{\text{train}}$

Randomly sample $\mathcal{T}_{\text{train}}$ from $\mathcal{D}_{\text{train}}$;
Transform the images in $\mathcal{T}_{\text{train}}$ by the data augmentation module;
Extract embedding vectors by the embedding network defined in (1);
Add label information to embedding vectors and construct the initial node features by (2);
for l in $\{1, 2, ..., L\}$ **do**
    Calculate edge features according to node features by (3) and (4);
    Calculate attention weights according to edge features by (5);
    Node features aggregation by (6);
    Node features update by (7);
end for
Calculate the embedding loss by (8), (9) and (10);
Calculate the classification loss by (11);
Calculate the mixed loss by (12);
Update the parameter set Θ by back-propagation according to (13).

In the training process, we feed $\mathcal{T}_{\text{train}}$ into the network and obtain the optimal network parameters by minimizing the mixed loss through gradient descent, as illustrated by the upper part of Fig. 1 and the pseudo code in Table III. In addition, we randomly sample 32 training tasks, i.e., $T = 32$, from the training set in each iteration (because each training task contains only one query sample, the batch size equals to 32) and adopt the Adam optimizer with the initial learning rate of 10^{-3}. Then, we reduce the learning rate exponentially by

$$\eta_\tau = \eta_0 \times g^{\frac{\tau}{\tau_b}} \tag{14}$$

where η_0 is the initial learning rate, $g = 0.9$ is the hyperparameter, τ is the iteration number, and $\tau_b = 5000$ is the number of training epoch.

In the testing process, we fix parameters of the MGA-Net and directly feed $\mathcal{T}_{\text{test}}$ into the network to obtain the classification result, as illustrated by the lower part of Fig. 1.

C. Classification Results of the MGA-Net

It should be noted that although the model parameters and the query set remain constant in the test stage, the classification accuracy varies with the support set. Therefore, we construct 1000 independent test task sets $\mathcal{T}_{\text{test}}$ in each experiment through random sampling. Table IV shows the classification results and Table V presents the average confusion matrices. By 17 s/15 q, we mean that the depression angles of the support set and the query set are 17° and 15°, respectively. It is observed that the average accuracy is above 90% and

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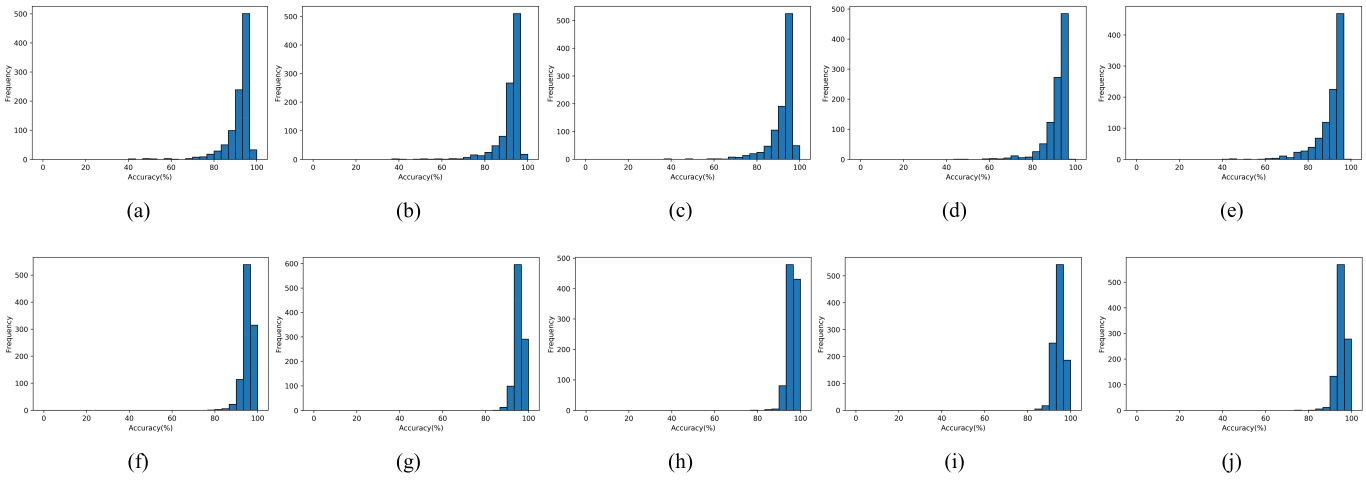


Fig. 7. Histograms of the test accuracy for 1-shot and 5-shot with various depression angles on the MSTAR dataset. (a) 15 s/15 q (1-shot). (b) 17 s/15 q (1-shot). (c) 17 s/17 q (1-shot). (d) 30 s/30 q (1-shot). (e) 17 s/30 q (1-shot). (f) 15 s/15 q (5-shot). (g) 17 s/15 q (5-shot). (h) 17 s/17 q (5-shot). (i) 30 s/30 q (5-shot). (j) 17 s/30 q (5-shot).

TABLE IV
EXPERIMENTAL RESULTS OF 3-WAY 1-SHOT AND 3-WAY 5-SHOT UNDER FIVE EXPERIMENTAL SETTINGS ON THE MSTAR DATASET

No.	Type		Accuracy (%)			Standard deviation
			Min	*Max*	*Mean*	
1	15s/15q	1-shot	40.75	97.93	91.66	0.004199
		5-shot	78.71	98.66	95.35	0.000543
2	17s/15q	1-shot	36.74	97.45	91.62	0.004271
		5-shot	84.79	98.66	95.60	0.000330
3	17s/17q	1-shot	36.69	97.76	91.82	0.004293
		5-shot	76.85	99.55	95.99	0.000418
4	30s/30q	1-shot	46.34	96.75	91.34	0.003172
		5-shot	84.09	99.07	94.54	0.000464
5	17s/30q	1-shot	43.32	96.75	90.39	0.004526
		5-shot	74.91	98.84	95.28	0.000480

95% for 1-shot and 5-shot settings, respectively, demonstrating the effectiveness of the MGA-Net. In addition, the minimum accuracy (which is below 40% for 1-shot) rises to above 74% for 5-shot.

Fig. 7 further presents the corresponding statistical histograms, where unrepresentative support sets (such as samples with drastic changes in the background and so on) yield to low accuracy or even invalidate the model, while representative support sets increase the accuracy to 97% and 99% for 1-shot and 5-shot, respectively. In addition, histograms for 5-shot have smaller variances and larger means than those for the 1-shot. Therefore, the choice of support set greatly affects the classification results; and increasing the number of shots will significantly boost the classification performance in accuracy and robustness. In addition, the MGA-Net has exhibited robustness to large depression angle variation and achieves high accuracy for 1-shot and 5-shot experiments.

D. Classification Results on the OpenSARShip Dataset

In order to further demonstrate the effectiveness and generalization ability of the MGA-Net, a series of experiments are carried out on the OpenSARShip dataset [53], which was generated by Shanghai Jiao Tong University in 2017. It contains 11346 image slices of 17 ship categories (one category corresponds to all the unknown ships) obtained by the Sentinel-1 satellite with VH and VV polarizations. Compared with the MSTAR dataset, this dataset has more complex background and larger shape variation even for the same category; thus, it demands higher model robustness.

Because most categories contain only a few samples, we first sort the original dataset by the number of samples. Then, we set the first three categories (which have the largest sample number) as the training set (i.e., Cargo, Tanner, and Other Type) and the following three categories as the test set (i.e., Tug, Fishing, and Dredging). Considering the different cardinal numbers, 800 and 80 samples are randomly selected from each category to form the training set and the test set, respectively. Sample images of the six categories are shown in Fig. 8, and the detailed information is shown in Table VI. Because the image size in the OpenSARShip dataset varies greatly, we crop or pad them to 128×128.

Similar to the MSTAR dataset, 1000 experiments are conducted for the 3-way 1-shot and 3-way 5-shot scenarios, respectively, and the minimum, maximum, mean, and standard deviation of the test accuracy are listed in Table VII. The average confusion matrix and the histogram are shown in Table VIII and Fig. 9, respectively. It is observed that the proposed MGA-Net achieves an accuracy of 70.18% and 80.20% in the 1-shot and 5-shot scenarios, respectively, demonstrating its effectiveness in few-shot SAR target classification tasks.

E. Ablation Study

First, we demonstrate the effectiveness of the data augmentation module and the mixed loss in MGA-Net by experiments

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TABLE V
AVERAGE CONFUSION MATRIX ON THE MSTAR DATASET

No.	Real \ Output	3-way 1-shot				3-way 5-shot			
		2S1	BRDM-2	ZSU-234	Acc. (%)	2S1	BRDM-2	ZSU-234	Acc. (%)
15s/15q	2S1	242.15	23.22	8.63	88.38	266.31	7.49	0.20	97.19
	BRDM-2	33.67	240.33	0	87.71	29.45	244.54	0	89.25
	ZSU-234	3.00	0	271.00	98.91	1.05	0	272.95	99.62
17s/15q	2S1	240.45	24.18	9.37	87.76	262.83	10.40	0.77	95.92
	BRDM-2	32.03	241.97	0	88.31	24.27	249.73	0	91.14
	ZSU-234	3.30	0	270.69	98.80	0.76	0	273.24	99.72
17s/17q	2S1	264.08	24.01	9.91	88.62	290.31	5.52	2.17	97.42
	BRDM-2	36.67	261.33	0	87.69	27.33	270.67	0	90.83
	ZSU-234	2.55	0	295.45	99.14	0.79	0	297.21	99.73
30s/30q	2S1	253.36	23.31	10.34	88.28	275.88	7.22	3.90	96.13
	BRDM-2	37.67	249.33	0	86.87	35.09	251.92	0	87.77
	ZSU-234	3.28	0	283.72	98.86	0.82	0	286.18	99.71
17s/30q	2S1	238.59	44.98	3.43	83.13	268.97	15.66	2.37	93.72
	BRDM-2	21.83	265.17	0	92.39	21.22	265.78	0	92.61
	ZSU-234	12.52	0	274.48	95.64	1.37	0	285.63	99.52

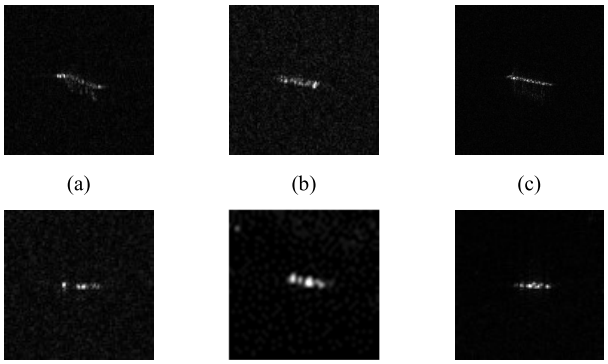


Fig. 8. Six categories of targets in the OpenSARShip dataset. (a) Cargo. (b) Tanker. (c) Other Type. (d) Tug. (e) Fishing. (f) Dredging.

TABLE VI
DIVISION OF THE OPENSARSHIP DATASET

	Target Class	Sample Number
Training set	Cargo Tanker Other Type	800
Test set	Tug Fishing Dredging	80

on the MSTAR dataset. Table IX shows the average accuracy of the 1000 experiments with (state = 1) and without (state = 0) data augmentation during training. After data augmentation, the average accuracy is increased by 15% and 6% for the 1-shot and 5-shot settings, respectively. Therefore, data augmentation facilitates model training and greatly improves the generalization performance.

TABLE VII
EXPERIMENTAL RESULTS OF 3-WAY 1-SHOT AND 3-WAY 5-SHOT ON THE OPENSARSHIP DATASET

Type	Accuracy (%)			Standard deviation
	Min	*Max*	*Mean*	
1-shot	13.75	82.92	70.18	0.015052
5-shot	56.67	85.42	80.20	0.000823

Table X shows the average accuracy of the 1000 experiments with only classification loss (state = 0) and mixed loss (state = 1). It is observed that the mixed loss increases the accuracy by 6%–10% and 2%–8% for the 1-shot and 5-shot settings, respectively. In order to explain the effectiveness of the mixed loss explicitly, we utilize the t-distributed stochastic neighbor embedding (t-SNE) [54] to visualize the embedding vectors. As shown in Fig. 10, the mixed loss significantly enhances the separability of embedding vectors (especially for 2S1 and BRDM-2), which is conducive to the category inference of GAT. Furthermore, Fig. 11 shows the corresponding training loss, where the mixed loss facilitates faster convergence than the classification loss and reduces the training time by 40%.

In order to explain the attention mechanism in GAT explicitly, Fig. 12 visualizes the single attention weight of each query node to other nodes in the 3-way 1-shot setting on the MSTAR dataset. From left to right, the visualized attention weights for the first, second, and third layers together with the true labels are presented; and each row corresponds to a single test task.

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TABLE VIII
AVERAGE CONFUSION MATRIX ON THE OPENSARSHIP DATASET

	3-way 1-shot				3-way 5-shot			
Real \ Output	Tug	Fishing	Dredging	Acc. (%)	Tug	Fishing	Dredging	Acc. (%)
Tug	54.65	19.42	5.93	68.31	71.51	8.47	0.02	89.39
Fishing	10.84	65.59	3.57	81.99	11.71	68.25	0.04	85.31
Dredging	15.04	16.76	48.21	60.25	17.62	9.65	52.73	65.91

TABLE IX
COMPARISON OF EXPERIMENTAL RESULTS WITH AND WITHOUT DATA AUGMENTATION

	State	15s/15t	17s/15t	17s/17t	30s/30t	17s/30t
3-way 1-shot	0	77.78	76.39	78.62	76.14	69.16
	1	**91.66**	**91.62**	**91.82**	**91.34**	**90.39**
3-way 5-shot	0	88.49	88.88	90.05	91.53	87.02
	1	**95.35**	**95.60**	**95.99**	**95.54**	**95.28**

TABLE X
COMPARISON OF EXPERIMENTAL RESULTS WITH CLASSIFICATION LOSS AND MIXED LOSS

	State	15s/15t	17s/15t	17s/17t	30s/30t	17s/30t
3-way 1-shot	0	82.86	81.60	81.80	85.52	83.35
	1	**91.66**	**91.62**	**91.82**	**91.34**	**90.39**
3-way 5-shot	0	87.96	87.64	91.43	93.33	91.54
	1	**95.35**	**95.60**	**95.99**	**95.54**	**95.28**

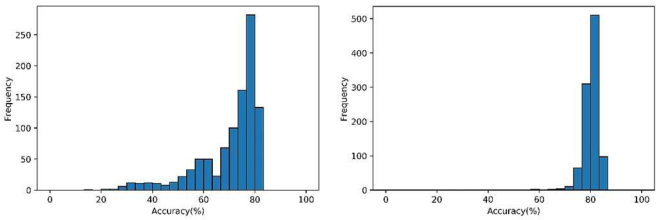


Fig. 9. Histograms of the test accuracy for (a) 3-way 1-shot (b) and 3-way 5-shot on the OpenSARShip dataset.

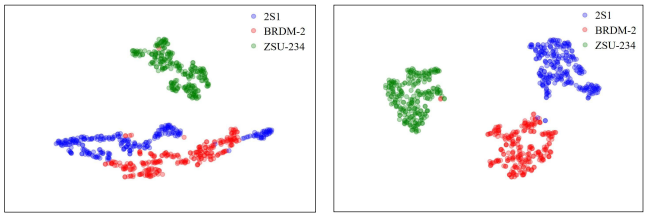


Fig. 10. t-SNE visualization of the embedding vector distribution under (a) classification loss and (b) mixed loss.

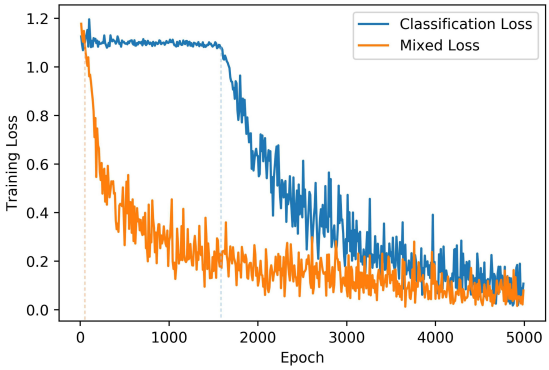


Fig. 11. Training loss with different loss functions.

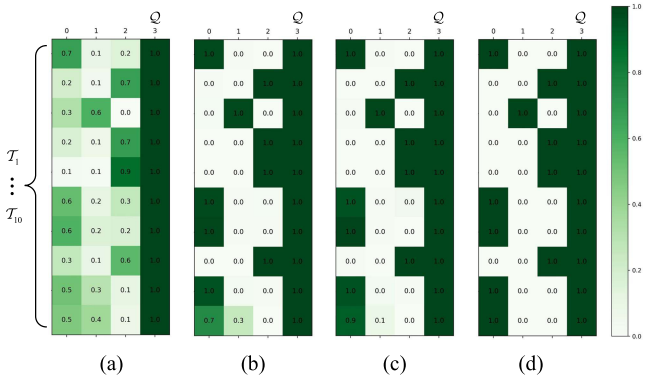


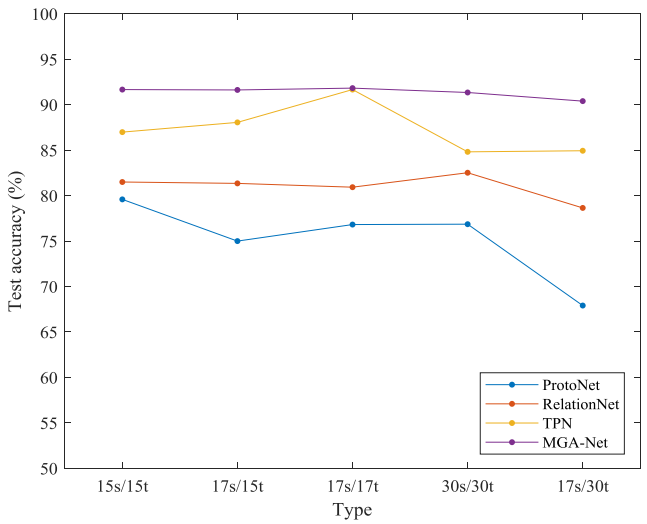
Fig. 12. Visualization of the attention weight for the (a) first layer, (b) second layer, (c) third layer, and (d) true label.

As an illustration, there are totally ten tasks and each task has three support samples labeled by 0, 1, and 2, respectively. In addition, the query sample is labeled by 3. Stronger color intensity in the grid represents larger values of the attention weight. For convenience, the self-attention weight of the query node is set to be 1.

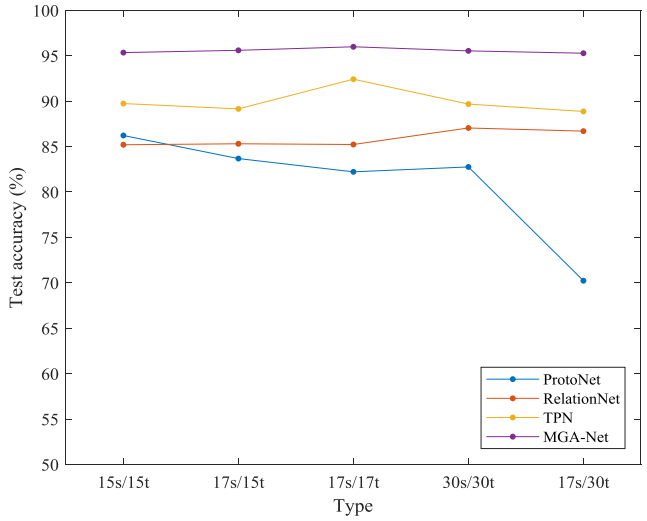
It is observed that for the same task, the attention weights of the query node to the support nodes are similar to the true label the query node belongs to, i.e., the query node mainly focuses on the support node with the same category and receives information from that node for feature update. Therefore, the attention mechanism enhances effective information and alleviates the interference from other nodes with different categories on the whole information flow.

Meanwhile, visualization of the attention weights of layer 1, 2, and 3 shows that deep layers in GAT enhances the attention of nodes, making it approachable to real-world situations and generate more accurate classification results.

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(a)



(b)

Fig. 13. Test accuracy curves of different models in (a) 3-way 1-shot and (b) 3-way 5-shot scenarios on the MSTAR dataset.

In addition, targeted information propagation gradually enhances the dissimilarity of node features, thereby avoids over-smooth due to the gradual similarity of node features in deep layers.

F. Contrast Experiments

Based on the same backbone, i.e., the convolution network (ConvNet), the average test accuracies of the MGA-Net and other classical few-shot classification models, such as the ProtoNet [39], the RelationNet [41], and Transductive Propagation Network (TPN) [40], are compared with the MSTAR and the OpenSARShip datasets, respectively, as shown in Tables XI and XII. The corresponding test accuracy curves of Table XI are shown in Fig. 13. It is observed that the MGA-Net obtains higher classification accuracy and higher robustness in various scenarios. In particular, for the 17 s/30 t experimental setup, the average accuracy of the ProtoNet decreases

TABLE XI
PERFORMANCE COMPARISONS ON THE MSTAR DATASET

	Model	15s/15t	17s/15t	17s/17t	30s/30t	17s/30t
3-way 1-shot	ProtoNet [39]	79.58	74.99	76.81	76.85	67.90
	RelationNet [41]	81.49	81.34	80.92	82.51	78.64
	TPN [40]	86.98	88.05	91.66	84.81	84.93
	MGA-Net (Ours)	**91.66**	**91.62**	**91.82**	**91.34**	**90.39**
3-way 5-shot	ProtoNet [39]	86.22	83.68	82.22	82.76	70.23
	RelationNet [41]	85.20	85.31	85.23	87.05	86.70
	TPN [40]	89.74	89.15	92.42	89.68	88.87
	MGA-Net (Ours)	**95.35**	**95.60**	**95.99**	**95.54**	**95.28**

TABLE XII
PERFORMANCE COMPARISONS ON THE OPENSARSHIP DATASET

Model	3-way 1-shot	3-way 5-shot
ProtoNet [39]	54.99	75.30
RelationNet [41]	55.67	74.17
TPN [40]	63.45	76.45
MGA-Net (Ours)	**70.18**	**80.20**

significantly. In this scenario, the support and query sets of the same target category cannot form an effective cluster in the embedding space, and the Euclidean distance metric adopted in the ProtoNet generates large errors. On the contrary, the MGA-Net exhibits robustness to depression angle variation and achieves high accuracy. For the OpenSARShip dataset, Table XII demonstrates that the classification accuracy of the MGA-Net is improved by 6.73%–15.19% and 3.75%–6.03% in 3-way 1-shot and 3-way 5-shot scenarios, respectively.

IV. CONCLUSION

In order to achieve robust few-shot SAR automatic target classification, this article proposes the MGA-Net, which augments the SAR images, maps them to low-dimensional embedding space by the embedding network, and outputs the labels by attention-enhanced GAT. In particular, the mixed loss is designed to gain strong similarity for intra-class and dissimilarity for inter-calss in the embedding space. Experimental results on the MSTAR and OpenSARShip datasets have shown that MGA-Net converges faster, achieves higher classification accuracy, and exhibits more robustness to large depression angle variation than other methods.

Future work will be focused on designing the robust few-shot SAR automatic classification frameworks under complex conditions such as target deformation and strong jamming and on exploiting sequential features of the SAR image series for robust few-shot classification.

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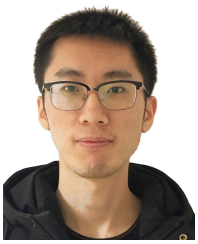
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